Exp: 06

Write the python program for Vacuum Cleaner Problem.

Aim:

To write a python program for vacuum cleaner problem.

Objective:

To understand the concept of a simple reflex agent that perceives its environment (room status) and acts (suck/move) based on condition-action rules.

Algorithm:

Step 1: Create a data structure to represent the state of the problem, including the current position of the vacuum cleaner and the state of each square in the room (clean or dirty).

Step 2: Create a data structure to represent the state space of the problem, including the initial state, the goal state, and the possible transitions between states.

Step 3: Use a search algorithm, such as depth-first search or breadth-first search, to explore the state space and find a solution.

Step 4: At each step of the search, generate all possible successor states from the current state by applying one of the allowed actions (move up, down, left, right, or clean).

Step 5: Check if the successor state is valid (i.e., the vacuum cleaner cannot move outside the room, and it can only clean dirty squares).

Step 6: If the successor state is valid, add it to the search tree and continue the search.

Step 7: If the successor state is the goal state, return the path from the initial state to the goal state.

Step 8: If there are no more possible successor states to explore, backtrack to the previous state and try another possible action.

Program:

```
def vacuum_cleaner():
    rooms = {'A': 'Dirty', 'B': 'Dirty'}
    current = 'A'
```

```
    while True:
        print("\nCurrent Location:", current)
        print("Room Status:", rooms)

        if rooms[current] == 'Dirty':
            print("Action: Clean")
            rooms[current] = 'Clean'
        else:
            print("Action: Move")

            if current == 'A':
                current = 'B'
            else:
                current = 'A'

        if rooms['A'] == 'Clean' and rooms['B'] == 'Clean':
            print("\nBoth rooms are clean.")
            break

vacuum_cleaner()
```

Output:

```
Current Location: A
Room Status: {'A': 'Dirty', 'B': 'Dirty'}
Action: Clean

Current Location: A
Room Status: {'A': 'Clean', 'B': 'Dirty'}
Action: Move

Current Location: B
Room Status: {'A': 'Clean', 'B': 'Dirty'}
```

Action: Clean

Both rooms are clean.

Result:

Hence, the simple python program for Vacuum Cleaner Problem was done